

Assignment Report

Muhammad Mahdi Jaffery

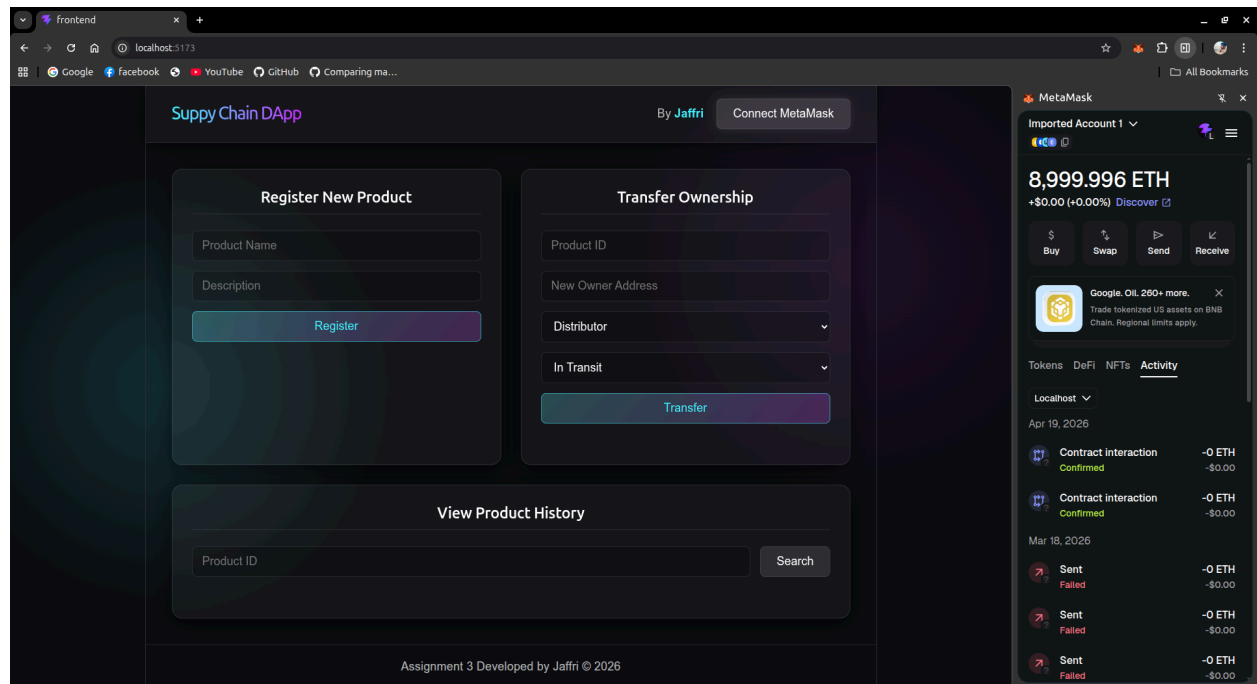
22L-6990
BS(CS)-8B

Compiling the hardhat

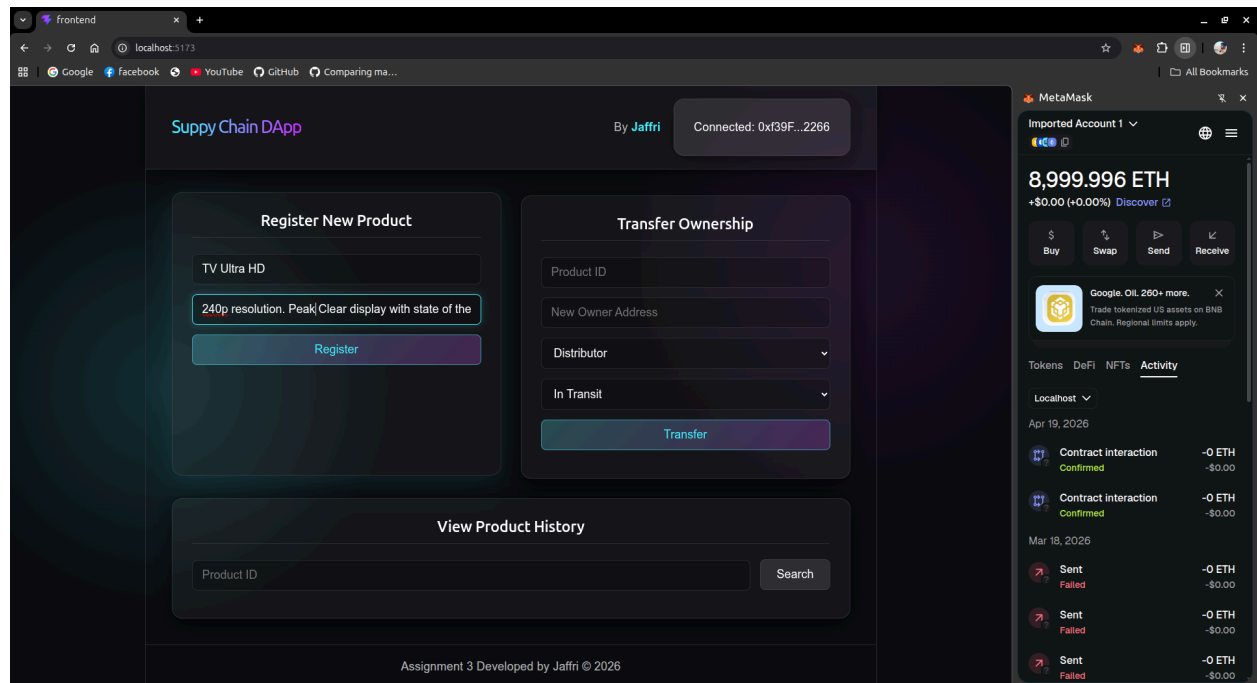
```
jaffri@Jaffri: ~/Desktop/8th Semester/BlockChain/Asm2/smart-contracts
jaffri@Jaffri:~/Desktop/8th Semester/BlockChain/Assignment 3$ cd ..
jaffri@Jaffri:~/Desktop/8th Semester/BlockChain$ cd Asm2/
jaffri@Jaffri:~/Desktop/8th Semester/BlockChain/Asm2$ ls
bcasm2.pdf  frontend  smart-contracts
jaffri@Jaffri:~/Desktop/8th Semester/BlockChain/Asm2$ cd smart-contracts/
jaffri@Jaffri:~/Desktop/8th Semester/BlockChain/Asm2/smart-contracts$ npx hardhat node
(node:67963) ExperimentalWarning: CommonJS module /home/jaffri/.npm/versions/node/v23.3.0/lib/node_modules/npm/node_modules/debug/src/node.js is loading ES Module /home/jaffri/.npm/versions/node/v23.3.0/lib/node_modules/npm/node_modules/supports-color/index.js using require().
Support for loading ES Module in require() is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
WARNING: You are currently using Node.js v23.3.0, which is not supported by Hardhat. This can lead to unexpected behavior. See https://v2.hardhat.org/nodejs-versions

(node:67976) ExperimentalWarning: CommonJS module /home/jaffri/Desktop/8th Semester/BlockChain/Asm2/smart-contracts/node_modules/hardhat/internal/core/config/config-loading.js is loading ES Module /home/jaffri/Desktop/8th Semester/BlockChain/Asm2/smart-contracts/hardhat.config.js using require().
Support for loading ES Module in require() is an experimental feature and might
```

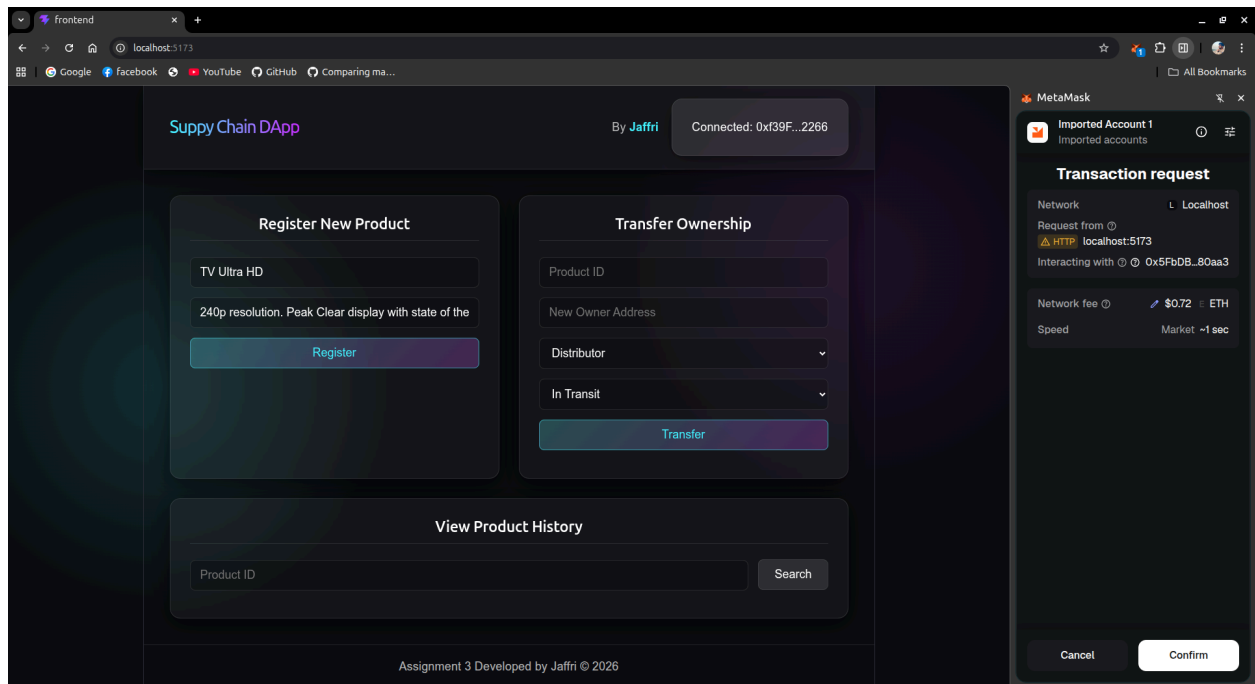
Switching to Manufacturer Account



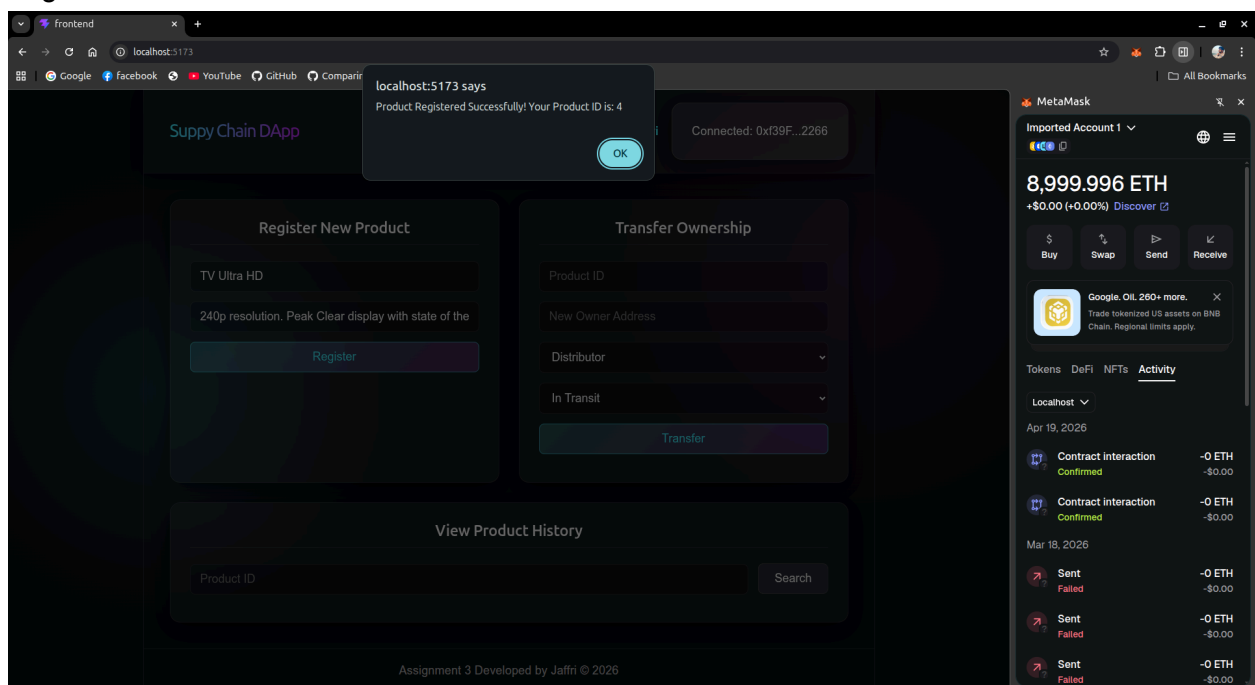
Registering Product



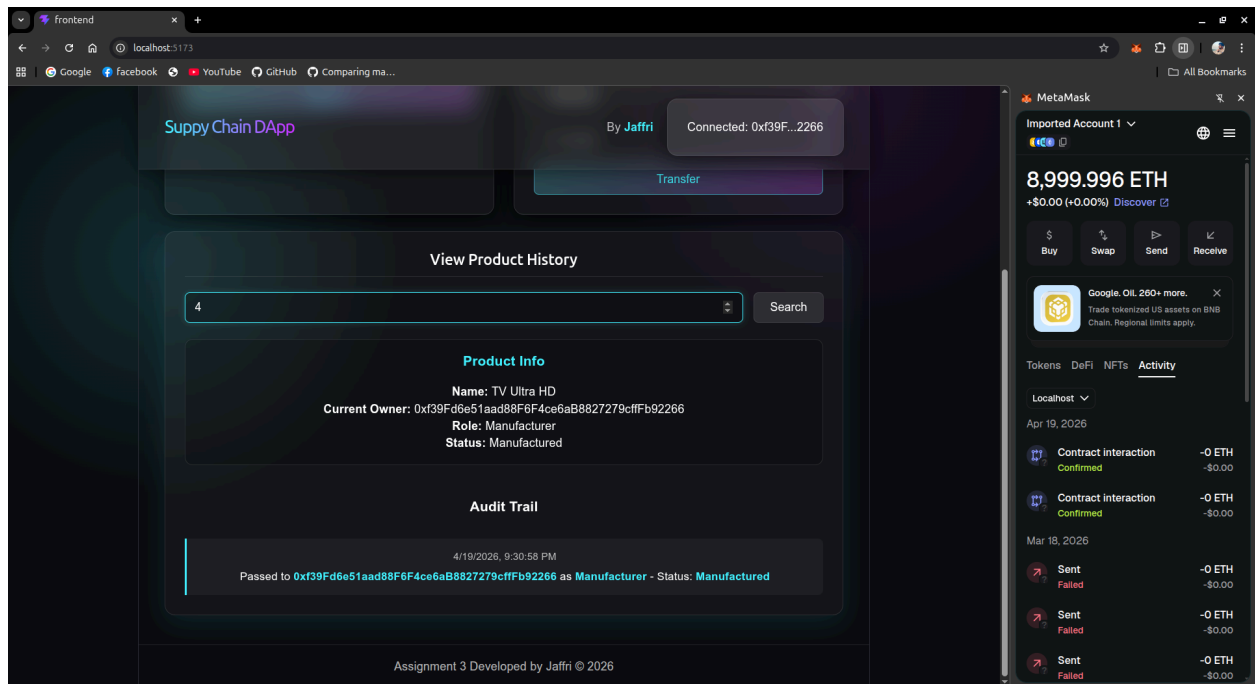
Product Registration request



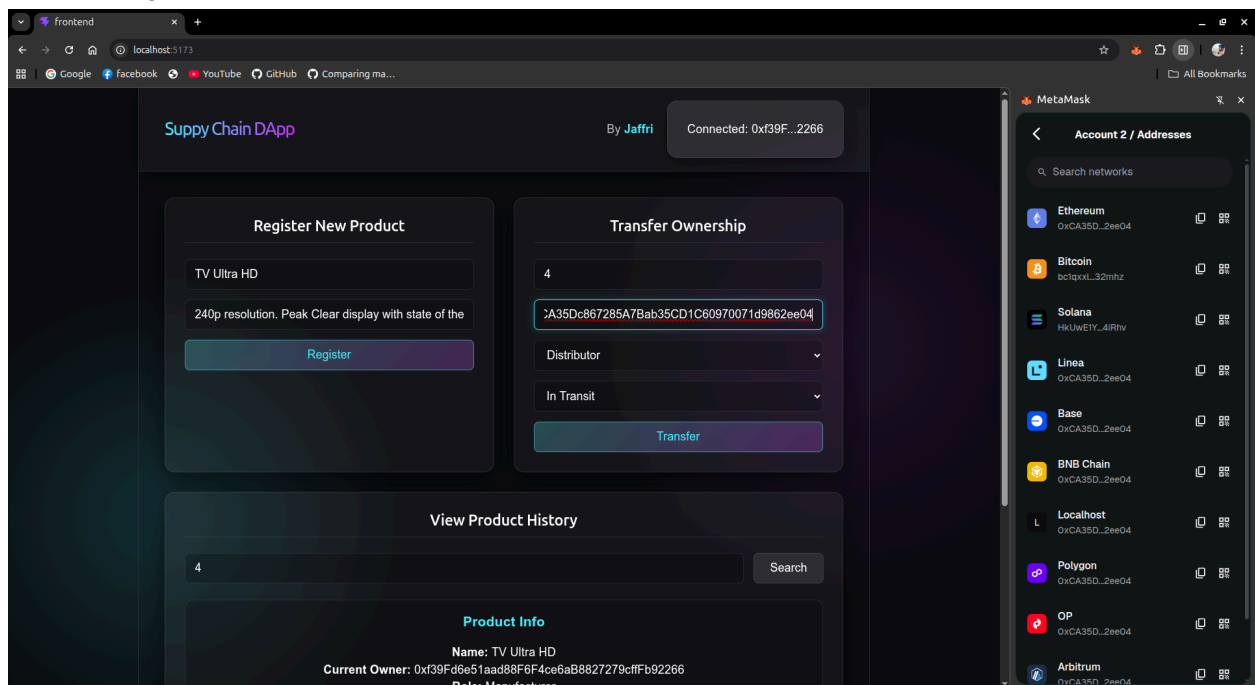
Registration Successful



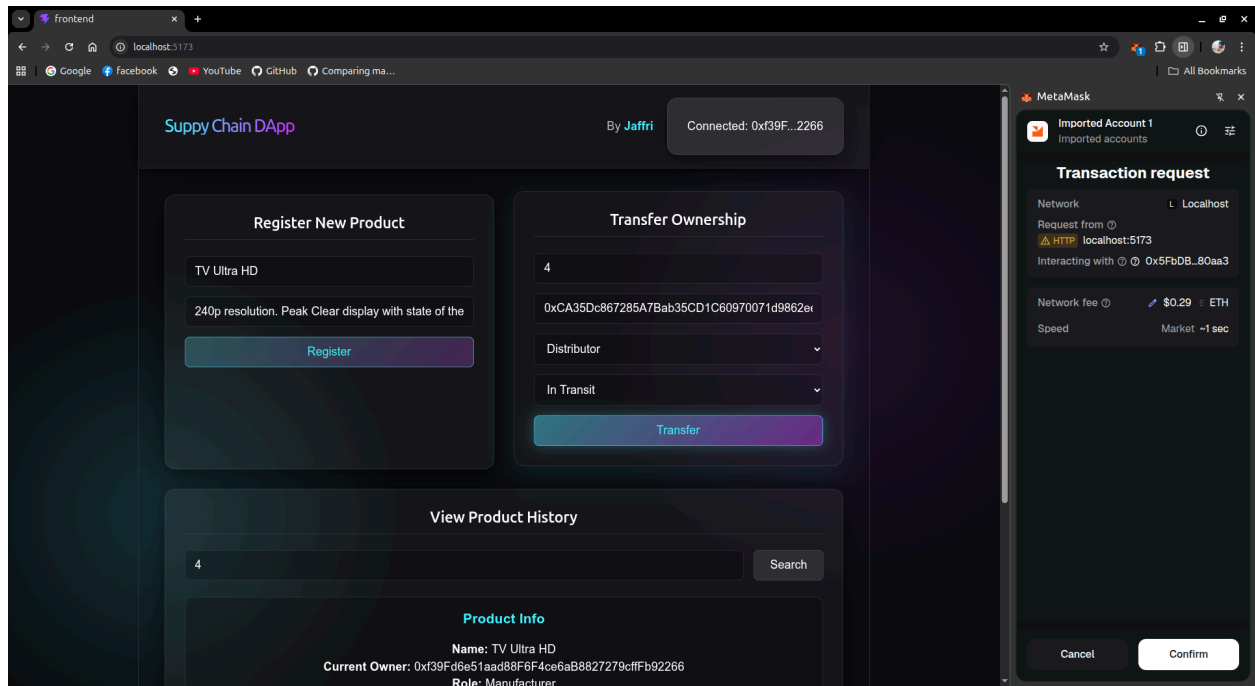
Product Trail



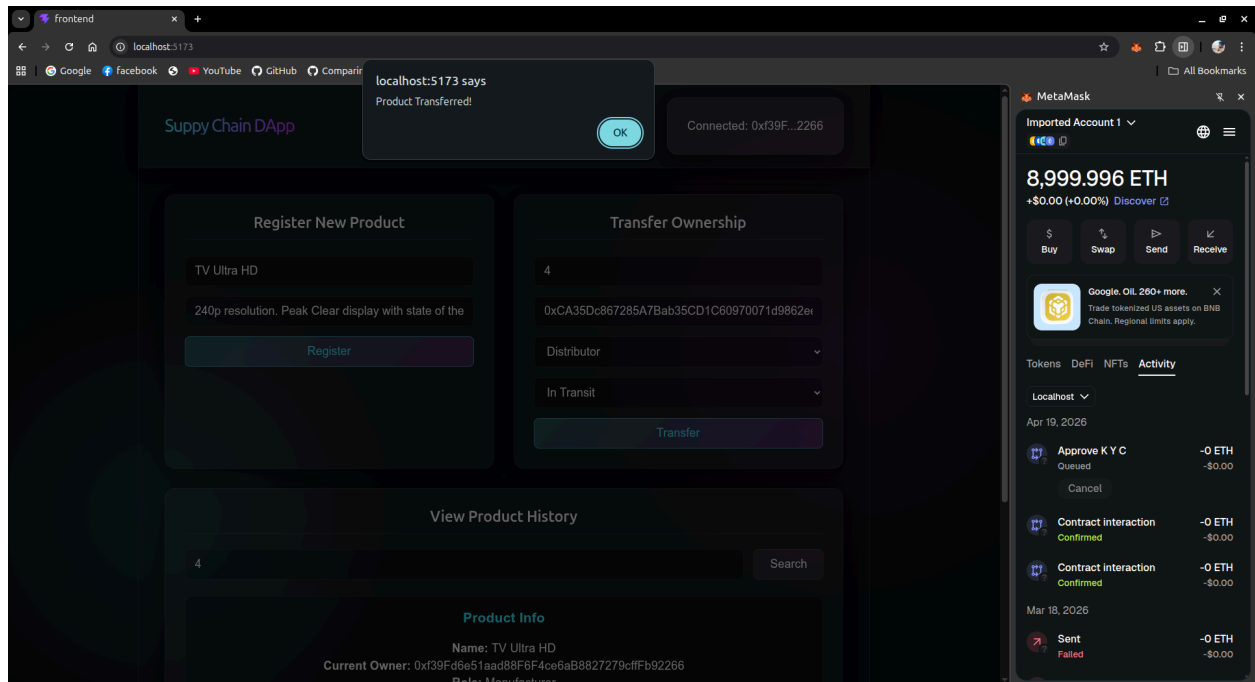
Transferring product ownership from Account 1 to Account 2



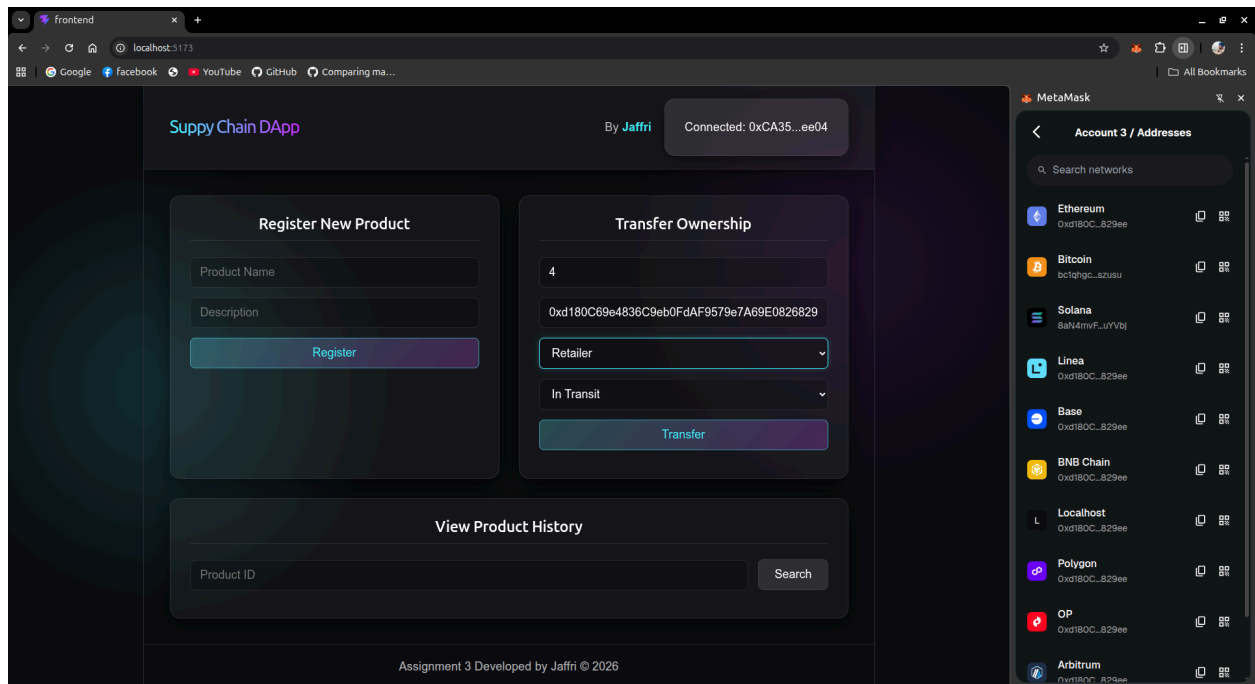
Transfer request



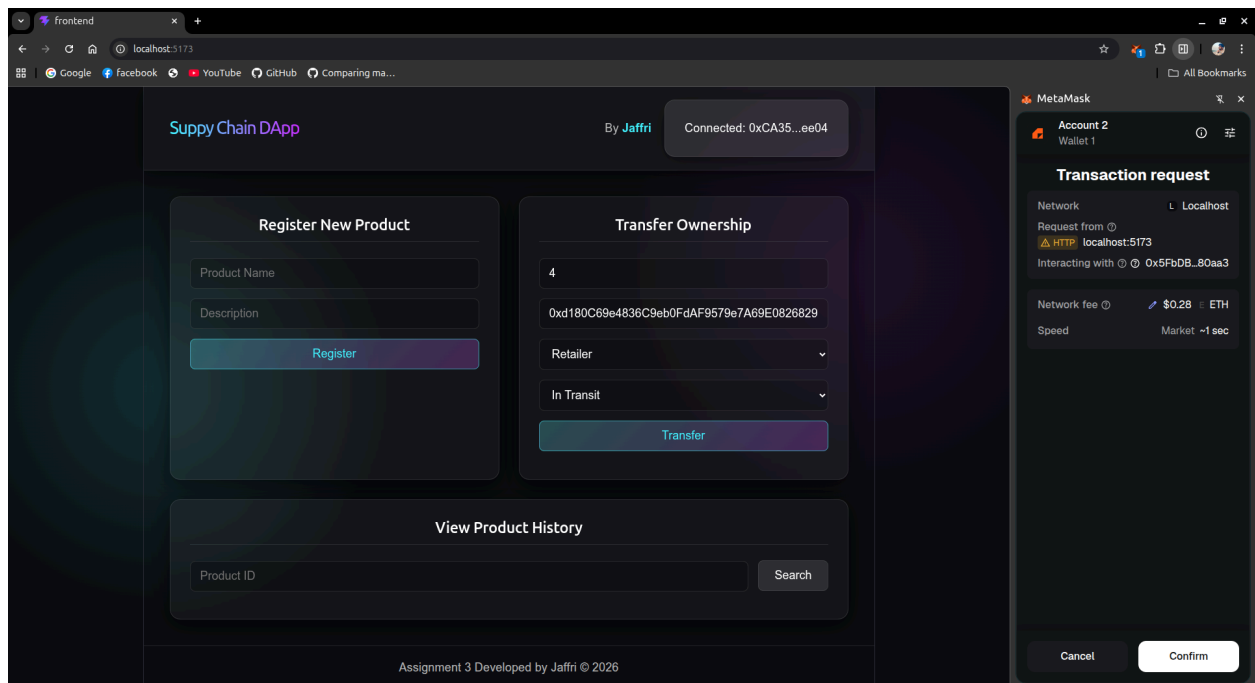
Transfer successful



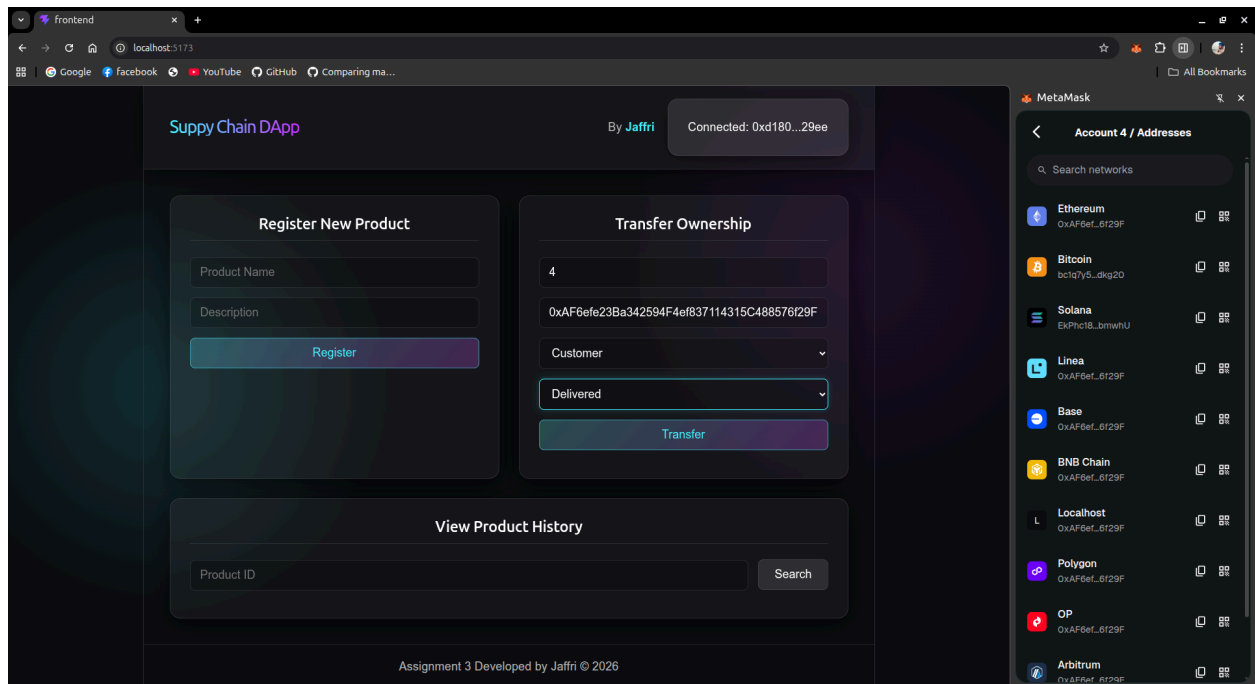
Transferring ownership from account 2 to account 3



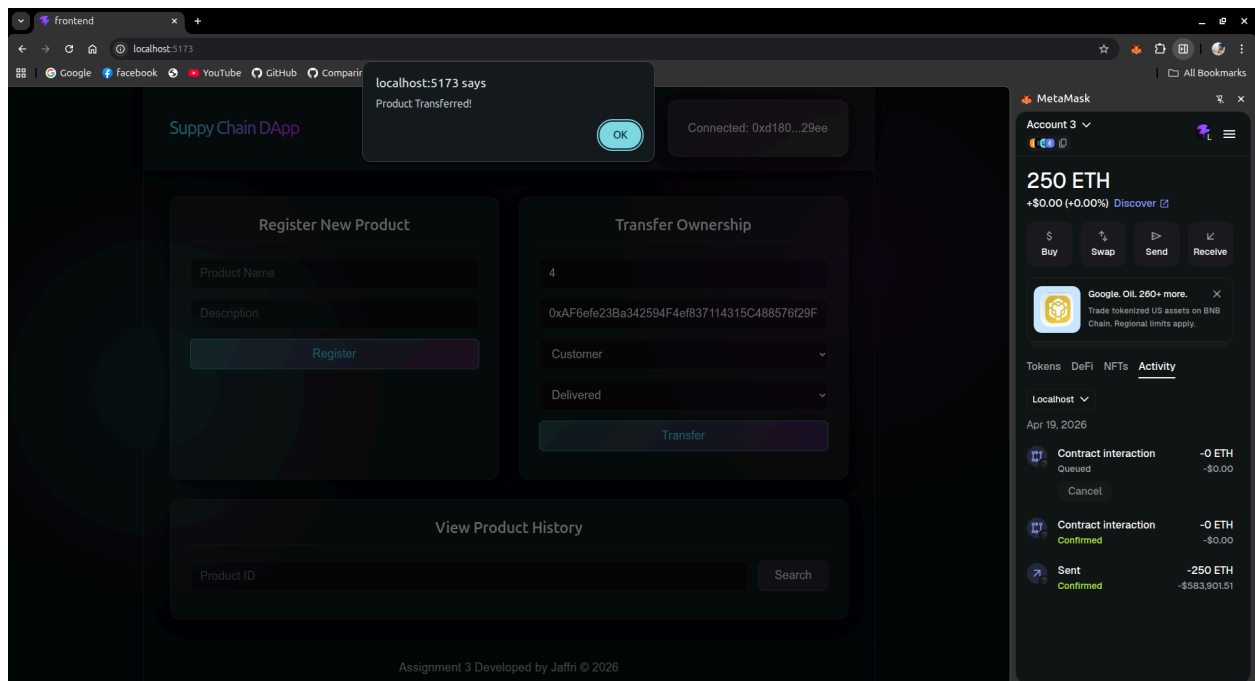
Transfer request



Transferring from account 3 to 4



Transfer successful



Product Trail

frontend

localhost:5173

GoogleFacebookYouTubeGitHubComparing ma...

Supply Chain DApp

By Jaffri

Connected: 0xd180...29ee

4

Search

Product Info

Name: TV Ultra HD

Current Owner: 0xAF6efe23Ba342594F4ef837114315C488576f29F

Role: Customer

Status: Delivered

Audit Trail

4/19/2026, 9:30:58 PM

Passed to 0xf39Fd6e51aad88F64ce6aB8827279cFfFb92266 as Manufacturer - Status: **Manufactured**

4/19/2026, 9:31:55 PM

Passed to 0xCA35Dc867285A7Bab35CD1C60970071d9862ee04 as Distributor - Status: **In Transit**

4/19/2026, 9:32:51 PM

Passed to 0xd180C69e4836C9eb0FdaF9579e7A69E0826829ee as Retailer - Status: **In Transit**

4/19/2026, 9:33:48 PM

Passed to 0xAF6efe23Ba342594F4ef837114315C488576f29F as Customer - Status: **Delivered**

MetaMask

Account 3

250 ETH

+\$0.00 (+0.00%) Discover

Buy

Swap

Send

Receive

Google, Oil, 260+ more.

Trade tokenized US assets on BNB Chain. Regional limits apply.

Tokens

DeFi

NFTs

Activity

Localhost

Apr 19, 2026

Contract interaction

Confirmed

-0 ETH

-\$0.00

Contract interaction

Confirmed

-0 ETH

-\$0.00

Sent

Confirmed

-250 ETH

-\$583,901.51